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```
close all;
clear;
clc;
s=tf('s');
```

仿真模型构建

```
Rs = 5.70125;
Ld = 1.86152e-3;
Lq = 0.00194861;
Ts = 1/(1700000000 / 5665 / 2);
Psi = 0.0096;
%□波器系数
fi = 0.9;
%偏置修正
D_P_DC = -0.02;
D_N_DC = -0.02;

G_D = 1/(Rs+Ld*s);
G_Q = 1/(Rs+Lq*s);

G_D = c2d(1/(Rs+Ld*s) * exp(-1*Ts*s) * exp(-260e-9 * s) * 1/(120e-9 * s +
1), ...
Ts, 'zoh');
G_Q = c2d(1/(Rs+Lq*s) * exp(-1*Ts*s) * exp(-260e-9 * s) * 1/(120e-9 * s +
1), ...
Ts, 'zoh');

%□□□□序列段
Start = 950;
Stop = 1150;
```

D □正向方波

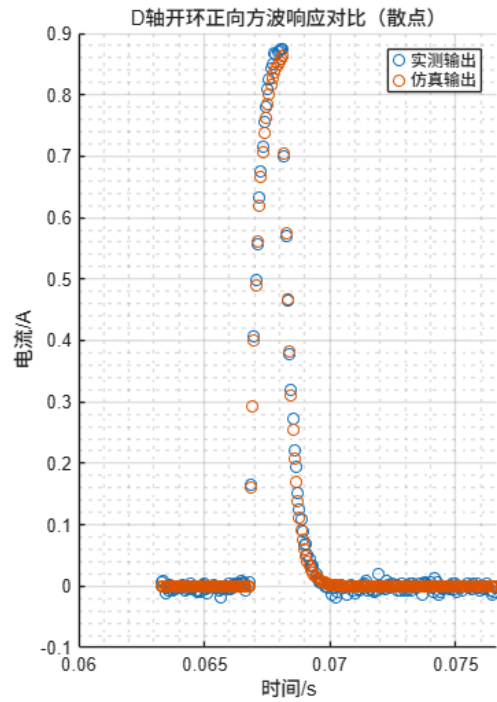
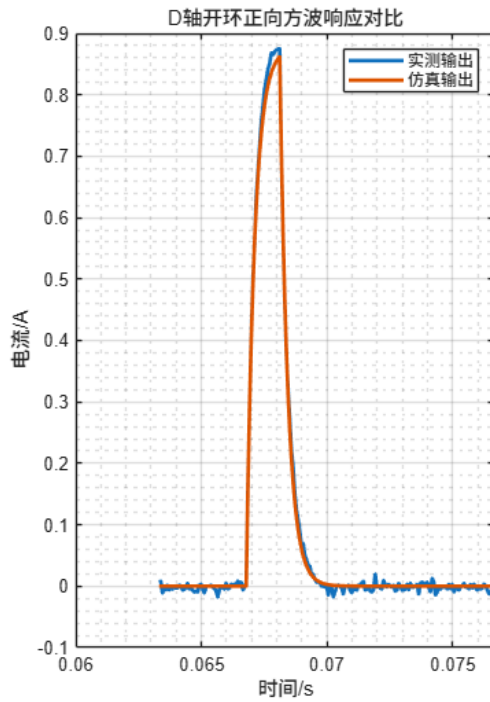
```
data = readmatrix("D_Step_P.txt", 'NumHeaderLines', 1);
theta_e = data(end-2046:end,2);
Id = data(end-2046:end,5);
we = zeros(length(Id),1);
we(1) = 0;
for i=2:length(we)
```

```

    we(i)=(theta_e(i)-theta_e(i-1))/Ts * 2 * pi / 360;
    if(abs(theta_e(i)-theta_e(i-1))>50)
        we(i)=we(i-1);
    end
end
we_lp = lowpass(we,fi);
input = data(end-2046:end,8) + we_lp .* (Lq .* data(end-2046:end,6));
output_rec = data(end-2046:end,5) + D_P_DC;
t = Ts:Ts:Ts*(length(input));
output_sim = lsim(G_D,input(1:end),t);
figure;
subplot(1,2,1);
plot(t(Start:Stop),output_rec(Start:Stop), ...
     t(Start:Stop),output_sim(Start:Stop),LineWidth=2);
subtitle('D □ □ □ 正向方波响□ □ 比');
legend('□ □ □ 出', '仿真□ 出');
grid on;
grid minor;
ylabel('□ 流/A');
xlabel('□ □ /s');
subplot(1,2,2);
scatter(t(Start:Stop),output_rec(Start:Stop));
hold on;
scatter(t(Start:Stop),output_sim(Start:Stop));
subtitle('D □ □ □ 正向方波响□ □ 比（散点）');
legend('□ □ □ 出', '仿真□ 出');
grid on;
grid minor;
ylabel('□ 流/A');
xlabel('□ □ /s');
fprintf('D □ □ □ 正向方波响□ : \n')
cov_sim_rec = cov(output_rec(Start:Stop), output_sim(Start:Stop));
cov_sim_rec=cov_sim_rec(1,2);
r=cov_sim_rec/(sqrt(var(output_rec(Start:Stop)))* ...
    sqrt(var(output_sim(Start:Stop))));
fprintf('相关系数 : %.5f%%\n',r*100);
r = 100 * (1 - norm(output_rec(Start:Stop) - output_sim(Start:Stop)) / ...
    norm(output_rec(Start:Stop) - mean(output_rec(Start:Stop))));
fprintf('□ 合□ 度 : %.5f%%\n\n',r);

D □ □ □ 正向方波响□ :
相关系数 : 99.95533%
□ 合□ 度 : 96.20818%

```



D 轴正向方波

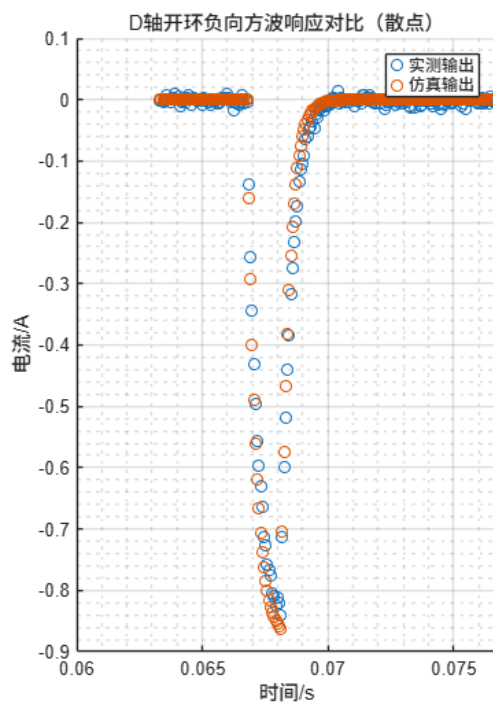
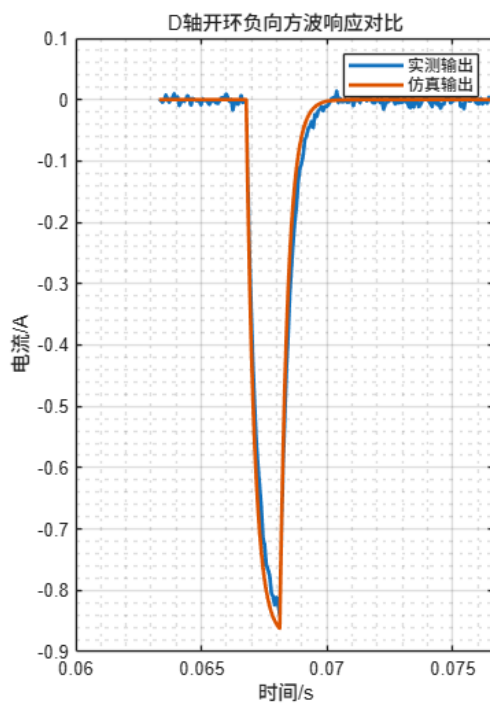
```
data = readmatrix("D_Step_N.txt", 'NumHeaderLines', 1);
theta_e = data(end-2046:end,2);
Id = data(end-2046:end,5);
we = zeros(length(Id),1);
we(1) = 0;
for i=2:length(we)
    we(i)=(theta_e(i)-theta_e(i-1))/Ts * 2 * pi / 360;
    if(abs(theta_e(i)-theta_e(i-1))>50)
        we(i)=we(i-1);
    end
end
we_lp = lowpass(we,fi);
input = data(end-2046:end,8) + we_lp .* (Lq .* data(end-2046:end,6));
output_rec = data(end-2046:end,5) + D_N_DC;
output_sim = lsim(G_D,input(1:end),t);
figure;
subplot(1,2,1);
plot(t(Start:Stop),output_rec(Start:Stop), ...
    t(Start:Stop),output_sim(Start:Stop),LineWidth=2);
subtitle('D 轴开□□向方波响□□比');
legend('□□□出', '仿真□出');
grid on;
grid minor;
ylabel('□流/A');
xlabel('□□/s');
subplot(1,2,2);
```

```

scatter(t(Start:Stop),output_rec(Start:Stop));
hold on;
scatter(t(Start:Stop),output_sim(Start:Stop));
subtitle('D 开 向方波响 比（散点）');
legend(' 出','仿真 出');
grid on;
grid minor;
ylabel(' 流/A');
xlabel(' /s');
fprintf('D 开 向方波响 : \n')
cov_sim_rec = cov(output_rec(Start:Stop), output_sim(Start:Stop));
cov_sim_rec=cov_sim_rec(1,2);
r=cov_sim_rec/(sqrt(var(output_rec(Start:Stop)))* ...
    sqrt(var(output_sim(Start:Stop))));
fprintf('相关系数 : %.5f%\n',r*100);
r = 100 * (1 - norm(output_rec(Start:Stop) - output_sim(Start:Stop)) / ...
    norm(output_rec(Start:Stop) - mean(output_rec(Start:Stop))));
fprintf(' 合 度 : %.5f%\n\n',r);

```

D 开 向方波响 :
 相关系数 : 99.59576%
 合 度 : 89.52477%



Q 正向方波

```

data = readmatrix("Q_Step_P.txt", 'NumHeaderLines', 1);
Uq = data(end-2046:end,9);
theta_e = data(end-2046:end,2);

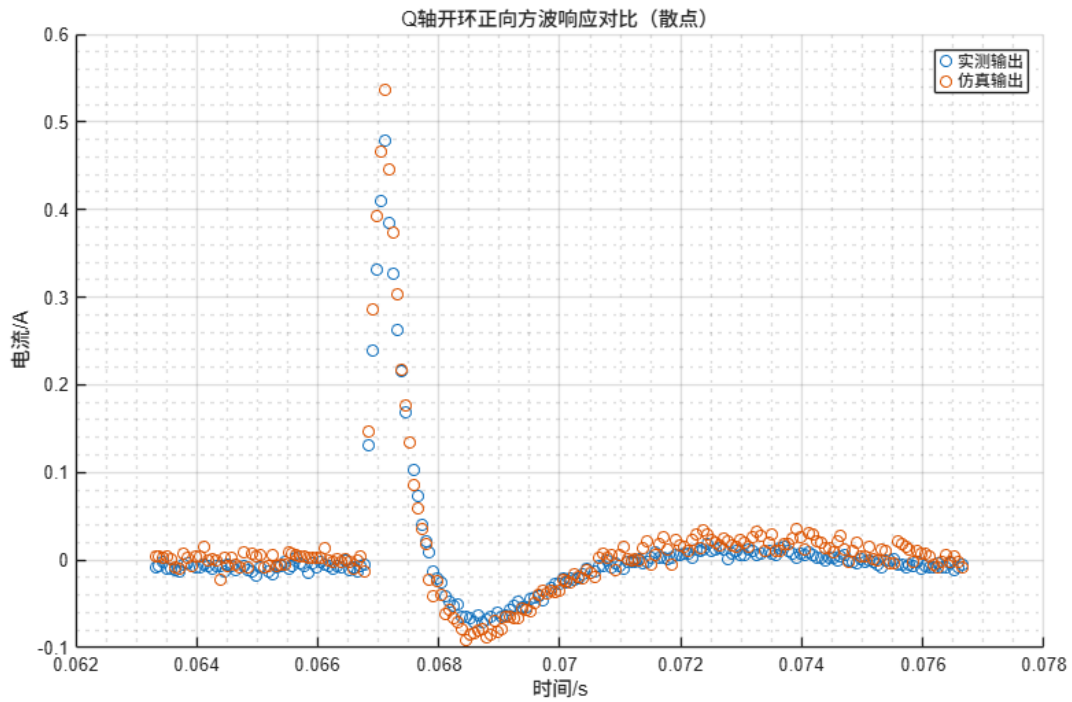
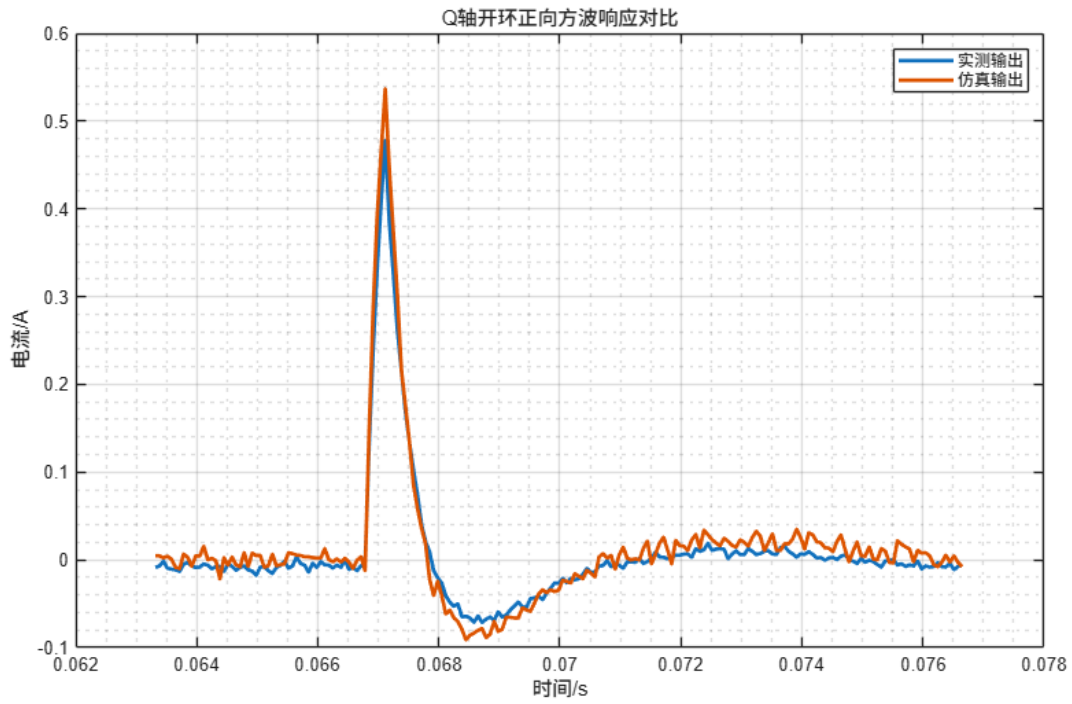
```

```

Id = data(end-2046:end,5);
we = zeros(length(Id),1);
we(1) = 0;
for i=2:length(we)
    we(i)=(theta_e(i)-theta_e(i-1))/Ts * 2 * pi / 360;
    if(abs(theta_e(i)-theta_e(i-1))>50)
        we(i)=we(i-1);
    end
end
we_lp = lowpass(we,fi);
input = Uq - we_lp .* (Ld * Id + Psi);
output_rec = data(end-2046:end,6);
output_sim = lsim(G_Q,input(1:end),t);
figure;
plot(t(Start:Stop),output_rec(Start:Stop), ...
     t(Start:Stop),output_sim(Start:Stop),LineWidth=2);
title('Q 开口正向方波响应比');
legend('开口出','仿真出');
grid on;
grid minor;
ylabel('电流/A');
xlabel('时间/s');
figure;
scatter(t(Start:Stop),output_rec(Start:Stop));
hold on;
scatter(t(Start:Stop),output_sim(Start:Stop));
title('Q 开口正向方波响应比（散点）');
legend('开口出','仿真出');
grid on;
grid minor;
ylabel('电流/A');
xlabel('时间/s');
fprintf('Q 开口正向方波响应：\n')
cov_sim_rec = cov(output_rec(Start:Stop), output_sim(Start:Stop));
cov_sim_rec=cov_sim_rec(1,2);
r=cov_sim_rec/(sqrt(var(output_rec(Start:Stop)))* ...
    sqrt(var(output_sim(Start:Stop))));
fprintf('相关系数：%.5f%%\n',r*100);
r = 100 * (1 - norm(output_rec(Start:Stop) - output_sim(Start:Stop)) / ...
    norm(output_rec(Start:Stop) - mean(output_rec(Start:Stop))));
fprintf('吻合度：%.5f%%\n\n',r);

Q 开口正向方波响应：
相关系数：99.08021%
吻合度：78.09775%

```



Q 轴 向

```
data = readmatrix("Q_Step_N.txt", 'NumHeaderLines', 1);
Uq = data(end-2046:end, 9);
```

```

theta_e = data(end-2046:end,2);
Id = data(end-2046:end,5);
we = zeros(length(Id),1);
we(1) = 0;
for i=2:length(we)
    we(i)=(theta_e(i)-theta_e(i-1))/Ts * 2 * pi / 360;
    if(abs(theta_e(i)-theta_e(i-1))>50)
        we(i)=we(i-1);
    end
end
we_lp = lowpass(we,fi);
input = Uq - we_lp .* (Ld * Id + Psi);
output_rec = data(end-2046:end,6);
output_sim = lsim(G_Q,input(1:end),t);
figure;
plot(t(Start:Stop),output_rec(Start:Stop), ...
    t(Start:Stop),output_sim(Start:Stop),LineWidth=2);
grid on;
grid minor;
title('Q 开 向方波响 比');
legend(' 出', '仿真 出');
ylabel(' 流/A');
xlabel(' /s');
figure;
scatter(t(Start:Stop),output_rec(Start:Stop));
hold on;
scatter(t(Start:Stop),output_sim(Start:Stop));
title('Q 开 向方波响 比（散点）');
legend(' 出', '仿真 出');
grid on;
grid minor;
ylabel(' 流/A');
xlabel(' /s');
fprintf('Q 开 向方波响 : \n')
cov_sim_rec = cov(output_rec(Start:Stop), output_sim(Start:Stop));
cov_sim_rec=cov_sim_rec(1,2);
r=cov_sim_rec/(sqrt(var(output_rec(Start:Stop)))* ...
    sqrt(var(output_sim(Start:Stop))));
fprintf('相关系数 : %.5f%%\n',r*100);
r = 100 * (1 - norm(output_rec(Start:Stop) - output_sim(Start:Stop)) / ...
    norm(output_rec(Start:Stop) - mean(output_rec(Start:Stop))));
fprintf('合 度 : %.5f%%\n\n',r);

Q 开 向方波响 :
相关系数 : 19.67163%
合 度 : -119.08079%

```

